1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

In breast cancer, TP53 mutation promotes EMT via upregulation of EMT transcription factors that suppress CDH1 (E-cadherin), driving remodeling of the tumor stroma and resulting in a high stromal-to-tumor ratio (TSR). This TSR–driven stromal expansion reflects EMT-associated desmoplasia and diminished epithelial integrity, and the prognostic impact of TSR is most pronounced when CDH1 is downregulated in TP53-mutant tumors. Clinically, TSR-high tumors with low CDH1 expression and TP53 mutation will have worse overall survival compared with TSR-high tumors with intact CDH1 and/or wild-type TP53. This relationship may be further modulated by immune-contexture, as reflected by Tier 2 modifiers such as LAG3 and another EMT/immunity-related gene, which could amplify or mitigate the TSR-associated risk by shaping the tumor microenvironment. The central test will be to demonstrate a significant TP53/CDH1–TSR interaction predicting OS in breast cancer, with replication across independent datasets and supportive mechanistic evidence from cell/in vivo models showing TP53 loss promotes EMT and reduces CDH1, contributing to stromal expansion and adverse outcome.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	CDH1, TP53
Primary endpoint	OS

2. Why This Gene — Not Others

Primary hypothesis gene(s)

CDH1: CDH1) and improve prognosis, but this epithelial rescue may fail in tumors with TP53 mutations that promote EMT and stromal remodeling; consequently, imaging-measured stromal expansion (high stromal-to-tumor ratio, TSR) does not uniformly translate to the same prognosis across molecular subtypes.

TP53: TP53 mutations that promote EMT and stromal remodeling; consequently, imaging-measured stromal expansion (high stromal-to-tumor ratio, TSR) does not uniformly translate to the same prognosis across molecular subtypes.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	Nanaomycin K	Supporting modifier from knowledge graph

Secondary	LAG3	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: quantify TSR from digitized histology,

Prediction 2: classify TP53 mutation status and CDH1 expression,

Prediction 3: test Cox models with TSR, TP53 status, CDH1, and interaction terms for OS,

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- Nanaomycin K: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- LAG3: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.